

# Likelihood-Ranked Synthetic Selection in Self-Consuming Diffusion: Minority Extinction Ordering on Imbalanced Mixtures

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## Abstract

Self-consuming training loops—retraining generative models on their own outputs—can induce *model collapse*, including loss of rare modes. We study unlabeled selection of synthetic samples by a denoising likelihood proxy in self-consuming DDPM training on imbalanced mixtures. Under matched real-mix ratio and matched synthetic budget, the first-generation minority-mass ratio  $r = m^{(1)}/m^{(0)}$  obeys

$$\text{bottom-}k > \text{random-}k > \text{top-}k$$

on a primary matrix of **eight** seed–imbalance cells (seeds  $\{0, 1, 2\}$  at  $\rho \in \{5, 10\}$  plus seeds  $\{3, 4\}$  at  $\rho=5$  only; means 0.57/0.36/0.17). Paired contrasts vs. random- $k$  show large effects (Cohen’s  $d_z \approx 1.0$ ) with Wilcoxon  $p \approx 0.042$  (Holm-adjusted  $p \approx 0.085$  for the two tests); we treat the ordering as supported by effect size and directional consistency, with small- $n$  caveats. Likelihood top- $k$  also concentrates majority mass relative to random- $k$ . A hypothesized “false stability” pattern—top- $k$  *improving* sliced  $W_2$  while minorities die—is **not observed**: at generation 1,  $\Delta W_2(\text{top} - \text{rand}) > 0$  on 8/8 cells (Wilcoxon  $p \approx 0.014$ ). Directional transfer checks on 8D GMM and Digits-PCA (3 seeds each) match the same order. We release configs, seeds, logs, and analysis code with the manuscript package.

## 1 Introduction

As synthetic data proliferates, generative models are increasingly trained on mixtures of real and model-generated samples. Iterated self-consumption can cause *model collapse*: progressive loss of diversity and forgetting of rare modes [Shumailov et al., 2023, Alemohammad et al., 2023, Gerstgrasser et al., 2024]. Mitigations include accumulating real data [Gerstgrasser et al., 2024], injecting fresh real samples, and *filtering* synthetic data [Feng et al., 2024, Cai et al., 2025].

When the only available score is the generator’s own likelihood (or denoising loss), how should one select synthetics? Feng et al. [2024] show that *task verifiers* for labeled synthetic answers can prevent collapse. Cai et al. [2025] filter “unrealistic” diffusion samples via latent probes and report aggregate FID/precision/recall. Alemohammad et al. [2023] observe that sampling bias toward high-quality generations drives a quality–diversity tradeoff. What is missing is a **matched-budget, mode-exact** causal contrast among likelihood ranks.

**What is new.** Prior work describes quality–diversity tradeoffs and filtering heuristics. We isolate an *interventional* selection policy  $\pi$  (top- $k$  vs. random- $k$  vs. bottom- $k$ ) at matched synthetic count and matched real-mix  $\alpha$ , and measure mode masses against known centers. The contribution is this controlled contrast and the resulting empirical minority-retention order—not a new named training algorithm or ImageNet-scale SOTA method.

## Contributions.

1. A controlled self-consuming DDPM protocol on imbalanced GMMs with exact mode masses.
2. Evidence for a likelihood-rank ordering of minority retention under matched  $k$  and  $\alpha$ .
3. Evidence that top- $k$  does *not* improve sliced  $W_2$  vs. random- $k$  while minorities decline (the opposite of a “better aggregate, worse minority” dissociation).
4. Directional transfer checks: 8D GMM and Digits-PCA with nearest class-mean occupancy.

## 2 Related work

**Model collapse.** Shumailov et al. [2023] document recursive training collapse and tail forgetting. Gerstgrasser et al. [2024] show accumulating real+synthetic data can avoid collapse. Alemohammad et al. [2023] (*MAD*) highlight quality-biased sampling.

**Verification and filtering.** Feng et al. [2024] argue verification of synthesized *labels* prevents collapse. Cai et al. [2025] (*LSF*) filter synthetic images using latent-space realism scores. Neither isolates unlabeled ELBO top- $k$  vs. random- $k$  with mode-grounded metrics on continuous mixtures.

**Temperature / diversity knobs.** Inference-time temperature methods [Xu et al., 2025, Li et al., 2026] reshape sampling diversity; we instead intervene on *training-set selection* inside self-consuming loops.

## 3 Method

### 3.1 Self-consuming loop

Let  $f^{(g)}$  be a DDPM trained on  $\mathcal{D}^{(g)}$ . Sample synthetic set  $\widehat{\mathcal{D}}^{(g+1)}$ . Build the next train set by mixing a fixed real buffer  $\mathcal{D}_{\text{real}}$  (fraction  $\alpha$ ) with selected synthetics.

### 3.2 Likelihood proxy and policies

Score  $s(x)$  is mean denoising MSE over a  $t$ -grid (lower = higher likelihood). At matched synthetic count  $n_{\text{syn}}$  and real fraction  $\alpha$ :

- **top- $k$  / bottom- $k$  / random- $k$ :** select a pool of size  $k = \max(n_{\text{syn}}, \lfloor k_{\text{frac}} N \rfloor)$  by lowest / highest / uniform scores, then draw  $n_{\text{syn}}$  from that pool and mix with  $\alpha$  real.
- **mix:** same  $\alpha$  real mix, but the synthetic portion is a uniform subsample of *all*  $N$  synthetics (no rank filter; no  $k$ -pool stage).
- **replace:** training set is pure synthetic (collapse reference).

Thus **mix**  $\neq$  **random- $k$** : random- $k$  still uses a size- $k$  subsample stage (matched to top/bottom); mix skips ranking entirely.

Table 1: Primary analysis cells (exactly eight seed–imbalance pairs). Seeds 3–4 were run only at  $\rho=5$  (replication);  $\rho=10$  uses seeds  $\{0, 1, 2\}$  only.

| $\rho \backslash$ seed | 0 | 1 | 2 | 3 | 4 |
|------------------------|---|---|---|---|---|
| 5                      | ✓ | ✓ | ✓ | ✓ | ✓ |
| 10                     | ✓ | ✓ | ✓ | — | — |

Table 2: Minority-mass ratio  $r = m^{(1)}/m^{(0)}$  on the primary eight-cell matrix (Table 1).

| Policy   | Mean $r$ | Std   | $n$ |
|----------|----------|-------|-----|
| bottom_k | 0.572    | 0.289 | 8   |
| rand_k   | 0.360    | 0.173 | 8   |
| mix      | 0.328    | 0.100 | 8   |
| top_k    | 0.172    | 0.078 | 8   |

### 3.3 Metrics

Nearest-mean mode assignment on known centers (GMM means, or Digits empirical class means in PCA space). Report minority-mean mass  $m$  (modes  $1 \dots K - 1$ ), majority mass (mode 0), and the ratio

$$r = \frac{m^{(1)}}{m^{(0)}},$$

which we call *retention* even when  $r > 1$  (minority mass can rise after one generation under bottom/random selection). Sliced  $W_2$ : average of squared 1D Wasserstein distances over **24** random projections; each comparison uses up to **400** synthetic and **400** held-out real points (same seed schedule as the run).

## 4 Experiments

### 4.1 Primary design (defines $n=8$ )

$K=4$  modes, tiny MLP denoiser,  $G=5$ ,  $\alpha=0.5$ ,  $k_{\text{frac}}=0.5$ . Logs: `w2_fs.jsonl` ( $n=228$  records).

### 4.2 Primary retention results

Table 2 averages  $r = m^{(1)}/m^{(0)}$  over the eight cells in Table 1.

At  $\rho=5$ , generation 1, all five seeds satisfy bottom  $>$  rand  $>$  top on minority mass. Paired retention contrasts vs. random- $k$  ( $n=8$ ): top–rand mean  $\Delta=-0.188$  (bootstrap 95% CI  $[-0.30, -0.06]$ ; Cohen’s  $d_z=-1.04$ ; Wilcoxon  $p \approx 0.042$ ); bottom–rand mean  $\Delta=+0.212$  (CI  $[0.08, 0.35]$ ;  $d_z=+1.00$ ;  $p \approx 0.042$ ). With two tests against the same baseline we apply Holm correction: both adjusted  $p \approx 0.085$  (above 0.05). We therefore emphasize **effect sizes**, CIs, and directional counts (7/8 top worse than rand; 6/8 bottom better) rather than uncorrected  $p$ -values alone. Top- $k$  also raises majority mass vs. random- $k$  at early generations.

### 4.3 Sliced $W_2$ (“false stability” check)

If top- $k$  were “falsely stable,” we would expect better (lower) sliced  $W_2$  than random- $k$  while minorities die. Instead, at generation 1,  $\Delta W_2(\text{top} - \text{rand})$  has mean  $+0.113$  (std 0.099), positive on

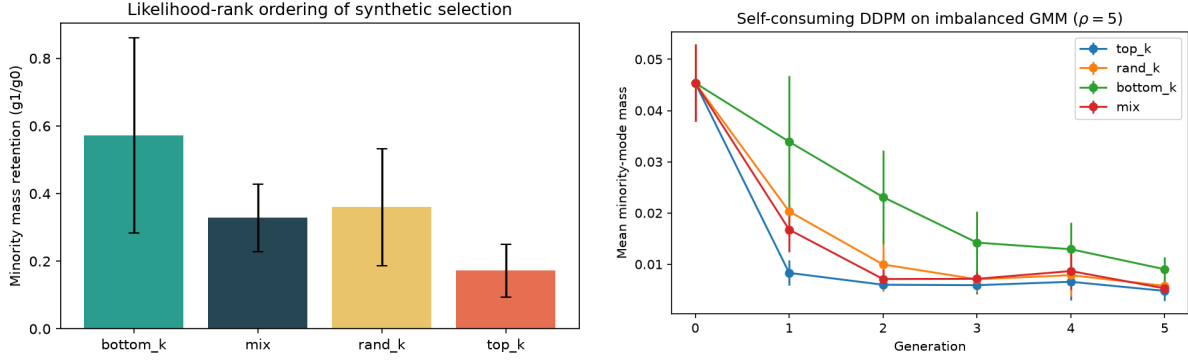


Figure 1: Left: retention by policy on the primary matrix. Right: minority mass vs. generation at  $\rho=5$ .

Table 3: Mean retention  $r$  by real-mix  $\alpha$  (two seeds). Order = bottom>rand>top on both seeds.

| $\alpha$ | bottom | rand | top  | Order holds |
|----------|--------|------|------|-------------|
| 0.25     | 0.25   | 0.17 | 0.05 | 1/2         |
| 0.50     | 0.62   | 0.49 | 0.24 | 2/2         |
| 0.75     | 1.34   | 0.77 | 0.94 | 0/2         |

**all eight** cells (Wilcoxon  $p \approx 0.014$ ). Means remain positive for generations 2–5 (6–7 of 8 cells; see `table_w2_top_minus_rand.csv`). We report this as **evidence against** a top- $k$   $W_2$  improvement under minority loss—not as a formal Neyman–Pearson “falsification” of a fully specified alternative.

#### 4.4 Real-mix sensitivity ( $\alpha$ )

Separate ablation (`w2_fs_alpha.jsonl`; seeds  $\{0, 1\}$ ,  $\rho=5$ ,  $G \leq 3$ ):

At  $\alpha \leq 0.5$ , top- $k$  remains worst; at  $\alpha=0.75$  the top vs. random gap reverses on average while bottom stays highest—an important scope condition ( $r > 1$  again means minority mass rose vs.  $g=0$ ).

#### 4.5 Transfer checks (directional; $n=3$ )

These are **not** powered significance tests; we report mean $\pm$ std and seed-wise order counts.

- **8D GMM** (`w2_fs_hd`): retention  $0.55 \pm 0.20$  /  $0.27 \pm 0.06$  /  $0.08 \pm 0.03$  (bottom/rand/top); order holds 3/3.
- **Digits-PCA** (`w2_fs_digits_pca`):  $1.49 \pm 0.11$  /  $1.11 \pm 0.08$  /  $0.68 \pm 0.17$ ; order holds 3/3. Ratios  $> 1$  mean minority mass increased from  $g=0$  to  $g=1$  under bottom/random (definition of  $r$  above).

An earlier Digits CVAE+classifier probe is inconclusive and unused for claims.

#### 4.6 Selection-pool size ( $k_{\text{frac}}$ )

Under  $k = \max(n_{\text{syn}}, \lfloor k_{\text{frac}} N \rfloor)$ ,  $k_{\text{frac}} \in \{0.25, 0.5\}$  are operationally equivalent when  $n_{\text{syn}}$  binds; at 0.75 top- $k$  remains worst.

## 5 Analysis

Likelihood top- $k$  preferentially retains points the model already explains well (majority modes), reducing minority inflow. Bottom- $k$  does the opposite relative to random. Aggregate sliced  $W_2$  is a poor monitor for minority survival: it moves with top- $k$  in the wrong direction for a “quality win.”

## 6 Limitations

- Toy 2D/8D GMMs and Digits-PCA with tiny networks; not ImageNet-scale.
- Primary Wilcoxon tests use only  $n=8$  pairs; uncorrected  $p\approx 0.042$  becomes Holm-adjusted  $p\approx 0.085$ . Inference relies on large  $d_z$ , CIs, and directional consistency.
- Transfer checks use three seeds without formal hypothesis tests.
- Digits CVAE transfer was inconclusive; Digits-PCA uses embeddings, not raw pixels.
- Effect strongest at moderate imbalance and early generations; high real mix ( $\alpha=0.75$ ) weakens top vs. random.
- Contribution is a controlled empirical contrast relative to MAD/LSF/Feng, not a new SOTA training method.

## 7 Conclusion

Unlabeled likelihood ranking of synthetic data in self-consuming diffusion induces a reproducible minority-retention *order* (bottom- $k > \text{random-}k > \text{top-}k$ ) under matched budgets on imbalanced mixtures, with large paired effect sizes despite small  $n$ . Top- $k$  does not improve sliced  $W_2$  relative to random- $k$  in our primary matrix. Preferring “most likely” synthetics is not a free lunch for minority survival.

## References

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## A Reproducibility and code

Cross-platform entry point: `python scripts/reproduce_w2.py` (also `scripts/reproduce_w2.sh` and Windows `scripts/reproduce_w2.ps1`). All numerical tables are produced from logged JSONL via analysis scripts (no hand-edited metrics). Experiment IDs: `w2_fs`, `w2_fs_hd`, `w2_fs_digits_pca` (supported); `w2_fs_digits` (CVAE; inconclusive). Source, configs, seeds, and logs: <https://github.com/muhammadalamzeb/likelihood-rank-self-consuming-diffusion> (commit recorded at submission time on the `master` branch).